

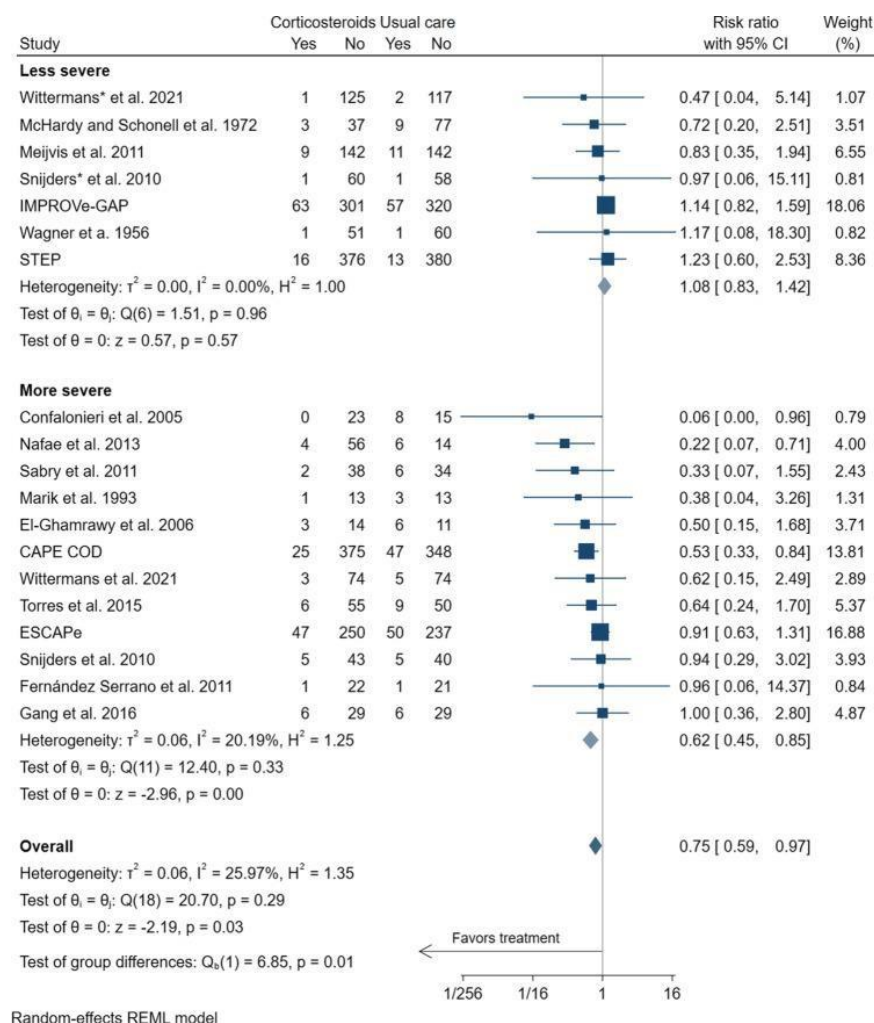
Evidence to Decision: Example of a Structured Evidence-to-Decision framework

This appendix provides a worked example of a structured Evidence-to-Decision (EtD) framework which explicitly considers each primary and secondary decision factors separately in moving from evidence to recommendations. The framework is presented in two formats produced using online platforms for digitally-structured guideline content: one using GRADEPro, and the second using MAGICapp.

Example: Consider a guideline panel is interested in producing practice guidance regarding the management of hospitalized adults with community-acquired pneumonia. The panel prioritizes the effects of short-term systemic corticosteroids, relative to no systemic corticosteroids, as a question of interest. Anticipating treatment effects are very likely to differ between patients with more severe pneumonia and less severe pneumonia, they construct separate PICOs for each patient group.

The research team supporting the panel searches for relevant literature, and identifies a recent systematic review of randomized controlled trials that compared short-term systemic corticosteroids versus standard care of placebo (without systemic corticosteroids) in adults hospitalized with suspected or probable community-acquired pneumonia.¹ The meta-analysis included all severities of disease and included a pre-specified sub-group analysis of more severe versus less severe pneumonia, where trials were defined as having patients with more severe disease if 50% or more of patients had severe pneumonia scores or were admitted to the intensive care unit at time of randomization (and were otherwise classified as having patients with less severe disease). The review identified 18 eligible studies including 4,661 patients. Results showed that corticosteroids probably reduce mortality in more severe pneumonia (relative risk [RR] 0.62, 95% confidence intervals [CI] 0.45 to 0.85, moderate certainty), and possibly have little or no effect in less severe pneumonia (RR 1.08, 95% CI 0.83 to 1.42, low certainty).

Figure 1: Forest plot for outcome mortality with subgroup analysis based on disease severity, from systematic review and meat-analysis of randomized trials comparing systemic corticosteroids to standard care or placebo in adults hospitalized with suspected or probable community acquired pneumonia. *Re-produced from Pitre J Gen Intern Med. 2023;38(11):2593-2606.*



The panel convenes to move from evidence to recommendations regarding the use of short-term systemic corticosteroids. They choose to focus on adults with less severe pneumonia for their first recommendation. To formulate recommendations, they consider primary factors from an individual patient perspective, including the magnitude of benefits, harms and burdens; the certainty of evidence; and the values and preferences of average patients with less severe pneumonia. They also consider relevant secondary factors from a population perspective, accounting for resource use, feasibility, acceptability and equity where applicable.

Panel discussions are structured around the EtD framework, where each factor is considered separately in moving from evidence to recommendations as summarized below. The EtD framework and associated evidence profile below are produced using GRADEpro, a digital platform for guideline development and dissemination.

QUESTION

Should short-term systemic corticosteroids versus no systemic corticosteroids be used for adults hospitalized with less severe community-acquired pneumonia?

POPULATION:	Immunocompetent adults aged 18 years or over and hospitalized with less severe community-acquired pneumonia (as defined by one or more validated pneumonia severity scoring systems or intensive care unit admission)
INTERVENTION:	Short-term systemic corticosteroids (7 days)
COMPARISON:	No systemic corticosteroids
MAIN OUTCOMES:	All-cause mortality; need for invasive mechanical ventilation; ventilator-free days; duration of hospitalization; time to symptom resolution; secondary infections; gastrointestinal bleeding.
SETTING:	In-patient hospital setting (emergency department, medical ward, and critical care units)
PERSPECTIVE:	The guideline provides clinical recommendations from an individual patient perspective. Guidance also takes into secondary population-level factors such as resource use, feasibility, acceptability and equity.
BACKGROUND:	Community-acquired pneumonia (CAP) is associated with severe pulmonary and systemic inflammation, resulting in impaired gas exchange, sepsis, and multi-organ failure. This dysregulated immune response is strongly associated with increased risk of re-hospitalization and death. Corticosteroids offer anti-inflammatory and immunomodulatory properties and are postulated to counteract the adverse effects observed in CAP. Randomized trials previously provided low to moderate certainty evidence that corticosteroids reduce time to symptom resolution, length of hospitalization and organ failure; effects on mortality were mixed (1,2). Two recent phase III, multi-center, double-blind randomized controlled trials (ESCAPE and CAPE COD) evaluating corticosteroids in hospitalized patients with CAP similarly showed mixed results (3,4). A clinical practice guideline published jointly by the American Thoracic Society and Infectious Disease Society of America in 2019, preceding publication of recent landmark trials, provided a conditional (weak) recommendation against routinely using corticosteroids in adults with severe CAP; this was based on moderate certainty evidence showing mixed effects on patient-important outcomes, lack of consistent definitions of disease severity used, and potential side-effects (5). More recently, a 2023 joint clinical practice guideline from the European Respiratory Society, European Society of Intensive Medicine and the Latin American Thoracic Association provided a conditional recommendation in favour of corticosteroids for severe CAP only where shock is present, but offered no guidance regarding less severe hospitalized CAP. No professional society has incorporated both recent trials into updated recommendations on corticosteroids for adults hospitalized with less severe CAP (6).
CONFLICT OF INTERESTS:	No relevant conflicts of interest.

ASSESSMENT

Problem Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ No ○ Probably no ○ Probably yes ● Yes ○ Varies ○ Don't know	Two recent phase III, multi-center, double-blind randomized controlled trials evaluating corticosteroids in hospitalized patients with CAP – CAPE COD (5) and ESCAPE (6) – have shown conflicting results: one demonstrated a significant 28-day survival benefit (5) and the second demonstrated no significant difference in 60-day mortality (6). No study evaluating extent of practice variation in corticosteroid administration for patients hospitalized with less severe CAP was identified.	The panel judged, based on their direct experience, that significant practice variation is likely to exist regarding use of corticosteroids for hospitalized patients with less severe CAP The panel agreed that the addition of two landmark trials to the growing body of evidence around corticosteroids in this context warranted new guidance. Both trials are larger than all previous smaller trials combined up to 2022. The panel therefore agreed that the role for corticosteroids in hospitalized patients with less severe CAP was a priority area warranting practice guidance.

Desirable Effects
How substantial are the desirable anticipated effects?

JUDGEMENT

- Trivial
- Small
- Moderate
- Large
- Varies
- Don't know

RESEARCH EVIDENCE

A literature review was conducted using OVID Medline and Embase to identify recent systematic reviews published since January 2015 addressing the prioritized guideline question. Of the identified reviews, one by Pitre and colleagues, published in the Journal of General Internal Medicine in April 2023, was used to inform the majority of outcomes prioritized by the panel (7).

Systemic corticosteroids result in 39 fewer patients requiring invasive mechanical ventilation (relative risk reduction of 46%) (high certainty), and probably prolong ventilator-free days by 2.9 days over a 90-day follow-up (moderate certainty). Corticosteroids may have a trivial effect on mortality (low certainty). Treatment probably reduces duration of hospitalization and time to symptom resolution (both moderate certainty).

Please refer to the GRADE evidence summary for full details. A more detailed evidence profile is provided following the EtD framework.

Outcomes	№ of participants (studies) Follow-up	Certainty of the evidence (GRADE)	Relative effect (95% CI)	Anticipated absolute effects* (95% CI)	
				Risk with no systemic corticosteroids	Risk difference with short-term systemic corticosteroids
All-cause mortality follow-up: 90 days	2246 (7 RCTs)	⊕⊕○○ Low	RR 1.08 (0.83 to 1.42)	Study population 81 per 1,000	7 more per 1,000 (14 fewer to 34 more)
Need for invasive mechanical ventilation follow-up: 90 days	2669 (9 RCTs)	⊕⊕⊕⊕ High	RR 0.56 (0.42 to 0.74)	Study population 89 per 1,000	39 fewer per 1,000 (52 fewer to 23 fewer)
Ventilator-free days follow-up: 90 days	630 (2 RCTs)	⊕⊕⊕○ Moderate	-	The mean ventilator-free days was 22.0 days	MD 2.9 days more (0.95 more to 4.85 more)
Duration of hospitalization follow-up: 90 days	2498 (8 RCTs)	⊕⊕⊕○ Moderate	-	The mean duration of hospitalization was 7.8 days	MD 1.1 days fewer (1.44 fewer to 0.72 fewer)
Time to symptom resolution	1322 (9 RCTs)	⊕⊕⊕○ Moderate	-	The mean time to symptom resolution was 5.4 days	MD 1.8 days fewer (2.45 fewer to 1.21 fewer)

ADDITIONAL CONSIDERATIONS

The panel pursued a stepwise process to establish minimal important difference (MID) thresholds for all critical and important outcomes of interest. After consulting clinical experts, they judged a systematic survey of studies evaluating patient values and preferences was likely to be low-yield, and conducting their own survey or focus groups with patients did not prove feasible. In the absence of such evidence, the panel relied on its own experiences engaging in shared decision-making with patients in judging the values and preferences of average patients, and accordingly their judgements regarding the threshold for important versus trivial effects.

The panel adopted a consensus-based approach to establish MIDs, and established thresholds for small, moderate and large effects for each prioritized outcome (though rated certainty in presence of an important versus a trivial effect):

- Mortality: 1% small but important effect, 1.5% moderate effect, 2% large effect.
- Need for invasive mechanical ventilation: 1% small but important effect, 1.5% moderate effect, 2% large effect.
- Ventilator-free days (mean difference): 1 day small but important effect, 2 days moderate effect, 3 days large effect.
- Duration of hospitalization (mean difference): 1 day small but important effect, 2 days moderate effect, 3 days large effect.
- Time to symptom resolution (mean difference): 2 days small but important effect, 3 days moderate effect, 4 days large effect.
- Secondary infections: 2% small but important effect, 3% moderate effect, 4% large effect.
- Gastrointestinal bleeding: 2% small but important effect, 3% moderate effect, 4% large effect.

Accordingly, the panel viewed treatment effects as being trivial for mortality, and small but important for duration of hospitalization and time to clinical cure.

The panel agreed that the moderate to large effects evident for invasive ventilation-related outcomes were in keeping with an overall large benefit despite lack of a survival benefit.

	<table><tr><td>follow-up: 90 days</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Secondary infections follow-up: 90 days</td><td>2730 (10 RCTs)</td><td>⊕⊕⊕○ Moderate</td><td>RR 1.09 (0.85 to 1.41)</td><td>Study population</td><td></td></tr><tr><td></td><td></td><td></td><td></td><td>84 per 1,000</td><td>8 more per 1,000 (13 fewer to 34 more)</td></tr><tr><td>Gastrointestinal bleeding follow-up: 90 days</td><td>3310 (11 RCTs)</td><td>⊕⊕⊕⊕ High</td><td>RR 0.95 (0.56 to 1.60)</td><td>Study population</td><td></td></tr><tr><td></td><td></td><td></td><td></td><td>18 per 1,000</td><td>1 fewer per 1,000 (8 fewer to 11 more)</td></tr></table>	follow-up: 90 days						Secondary infections follow-up: 90 days	2730 (10 RCTs)	⊕⊕⊕○ Moderate	RR 1.09 (0.85 to 1.41)	Study population						84 per 1,000	8 more per 1,000 (13 fewer to 34 more)	Gastrointestinal bleeding follow-up: 90 days	3310 (11 RCTs)	⊕⊕⊕⊕ High	RR 0.95 (0.56 to 1.60)	Study population						18 per 1,000	1 fewer per 1,000 (8 fewer to 11 more)	
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Undesirable Effects How substantial are the undesirable anticipated effects?																																
JUDGEMENT		RESEARCH EVIDENCE			ADDITIONAL CONSIDERATIONS																											
● Trivial ○ Small ○ Moderate ○ Large ○ Varies ○ Don't know		Systemic corticosteroids probably have a trivial effect on secondary infection (moderate certainty) and have a trivial effect on gastrointestinal bleeding (high certainty). Corticosteroids may result in 7 more deaths per 1000 treated patients. Please refer to GRADE evidence summary above and evidence profile table subsequently provided for full details.			The panel viewed treatment effects on secondary infections and gastrointestinal bleeding as being trivial. The panel similarly interpreted effects on mortality as being trivial, given a MID of 10 deaths per 1000 patients; they did, however, recognize that a reasonable proportion of patients may view this absolute increase in risk as small but important. Harms were deemed trivial overall.																											
Certainty of evidence What is the overall certainty of the evidence of effects?																																
JUDGEMENT		RESEARCH EVIDENCE			ADDITIONAL CONSIDERATIONS																											
○ Very low ● Low ○ Moderate ○ High ○ No included studies		For critically important outcomes, certainty of evidence ranged from low (for all-cause mortality) to high (for need for invasive mechanical ventilation). Overall certainty was therefore deemed to be low, in keeping with the lowest certainty rating among critical outcomes. For five of seven outcomes, evidence was downgraded for imprecision due to wide confidence intervals encompassing possibility of both important benefit and trivial effects. Certainty in mortality was rated down twice for imprecision, given confidence intervals encompassed the possibility of both important benefit and important harm while rating certainty in a trivial effect.																														
Values Is there important uncertainty about or variability in how much people value the main outcomes?																																
JUDGEMENT		RESEARCH EVIDENCE			ADDITIONAL CONSIDERATIONS																											
○ Important uncertainty or variability ● Possibly important uncertainty or variability ○ Probably no important uncertainty or variability ○ No important uncertainty or variability		A review of the COMET database yielded no existing core outcome data sets for patients hospitalized with CAP. A literature review regarding how patients hospitalized with CAP value the prioritized outcomes was not identified; an additional search for indirect evidence expanding to other forms of pneumonia or respiratory illness was de-prioritized due to feasibility considerations.			Possibly important uncertainty regarding patient values and preferences exists at two levels: one, the relative weight patients place on the prioritized outcomes; and two, the extent to which patients may value large benefits in avoiding invasive mechanical ventilation relative to trivial risk of harms including 90-day all-																											

		cause mortality, serious infections and gastrointestinal bleeding. The panel relied on its own judgments of what it perceived average well-informed patients would value when considering benefits, harms and burdens of treatment. The panel agreed that the majority of patients hospitalized with less severe CAP would be inclined to accept short-term therapy when faced with high certainty of moderately large or large benefit, and low certainty of trivial harm.
Balance of effects Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ Favors the comparison ○ Probably favors the comparison ○ Does not favor either the intervention or the comparison ● Probably favors the intervention ○ Favors the intervention ○ Varies ○ Don't know	High certainty evidence demonstrates a large reduction in need for invasive mechanical ventilation. Overall low certainty evidence demonstrates corticosteroids are possibly associated with trivial harm.	The panel placed greater value on a high certainty of large benefit observed in a non-mortality outcome, acknowledging low certainty of trivial harm. Significant uncertainty exists around harm outcomes, particularly all-cause mortality with a signal towards harm. The panel therefore agreed that the balance of effects probably, rather than definitely, favours the intervention.
Resources required How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ Large costs ○ Moderate costs ● Negligible costs and savings ○ Moderate savings ○ Large savings ○ Varies ○ Don't know	Systemic corticosteroids are inexpensive, easy to administer, and are readily available globally. Dexamethasone and prednisone are among the most commonly listed drugs on the national essential medicines list, available for use in over 95% of countries internationally (8). A seven-day course of corticosteroids has been shown to be an optimal regimen (7). Dexamethasone 6mg tablets costs approximately \$1.78 - \$1.90 USD per tablet (9); a 7-day course is estimated to cost a total of \$12.46-\$13.30 USD. While costs may vary somewhat across jurisdictions and based on available corticosteroid agents, dosages and formulations, the panel agreed they are likely to be negligible universally.	The systematic review by Pitre and colleagues suggests that 6mg of dexamethasone (or equivalent corticosteroid dosing for other agents) for a 7-day duration is an optimal treatment regimen. This was based on a non-linear dose-response relationship between corticosteroids and mortality (9). While impact on mortality among patients with less severe pneumonia was trivial, the panel agreed results may be extrapolated to suggest this to be a reasonable dose regimen for other outcomes where clear benefits and trivial harms were observed. The panel considered that a seven-day course of treatment is time-limited, and that the intervention is not associated with cumulative long-term costs. Panel members agreed that the cost of a one-week course of corticosteroids is likely to be affordable to most patients paying out-of-pocket. Moreover, the panel acknowledged that resources required, costs and barriers to access may be further mitigated in the in-patient setting, where corticosteroids are almost universally available, are often covered with governmental health insurance for in-patients, and require minimal resources to administer.
Certainty of evidence of required resources What is the certainty of the evidence of resource requirements (costs)?		

JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ Very low ○ Low ○ Moderate ● High ○ No included studies	The panel judged that, based on a large body of experience using systemic corticosteroids both for community-acquired pneumonia and for other respiratory illnesses, that there was high confidence in estimating resource requirements. Certainty regarding resource requirements and costs was further increased by review of crude costing information from drug formularies (9).	
Cost effectiveness Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ Favors the comparison ○ Probably favors the comparison ○ Does not favor either the intervention or the comparison ○ Probably favors the intervention ○ Favors the intervention ○ Varies ● No included studies	A literature review was conducted using PubMed with keywords "cost effectiveness", "corticosteroids" and "community acquired pneumonia" to identify studies evaluating cost effectiveness of systemic corticosteroids for community acquired pneumonia. One study by Pliakos and colleagues (10) evaluated the cost-effectiveness based on a decision analysis model which calculated deaths averted and incremental cost-effectiveness ratios with the addition of corticosteroids to antibiotics. Among hospitalized patients with Pneumonia Severity Index scores of I to III, corticosteroids resulted in additional costs of \$483,016 per death averted; only 29.9% of simulations demonstrated cost-effectiveness. Lack of cost-effectiveness for these patients was driven by very low mortality rates. No other recent cost-effectiveness analyses were identified.	Given a lack of survival benefit (and instead, a signal towards possible increased mortality) with corticosteroids, the panel viewed the results of the stated cost-effectiveness analysis, which is based on deaths averted, as less applicable to the current context. In the absence of other cost-effectiveness analyses, the panel acknowledged an absence of reliable cost-effectiveness data to base its judgments on.
Equity What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ Reduced ○ Probably reduced ○ Probably no impact ● Probably increased ○ Increased ○ Varies ○ Don't know	No research evidence was identified pertaining to equity. Systemic corticosteroids are anticipated to be accessible for hospitalized patients with less severe CAP across the vast majority of jurisdictions, including low and middle-income settings (8). Where patients may pay out of pocket for treatment, systemic corticosteroids are anticipated to contribute minimally to the total cost of care and are anticipated to be accessible from a financial perspective for patients across the vast majority of financial contexts.	The panel judged that recommending the use of systemic corticosteroids was unlikely to reduce, and may instead possibly increase, health equity. The panel did not identify plausible reasons for differences in the relative effectiveness of systemic corticosteroids for disadvantaged groups.
Acceptability Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ No ○ Probably no ○ Probably yes ● Yes ○ Varies ○ Don't know	No research evidence was identified pertaining to acceptability.	Systemic corticosteroids are easy to administer as an oral medication, and a short course of corticosteroids of 7 days is anticipated to be viewed favourably by the majority of patients. The panel, including four patient partners, agreed that most patients hospitalized with less severe CAP would be inclined to accept short-term systemic corticosteroids when faced with moderate or high certainty of moderately large or large potential benefit; and low, moderate or high certainty of trivial risk of harms. The panel therefore judged that the intervention would be acceptable to all or almost all patients.

		The panel also considered acceptability issues for health care providers involved in the administration of the medication. Systemic corticosteroids were not perceived to pose any major challenges in terms of drug administration.
Feasibility Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
○ No ○ Probably no ○ Probably yes ● Yes ○ Varies ○ Don't know	Systemic corticosteroids are widely available (8) and are easy to administer as oral medications. Treatment course is short and time-limited at 7 days. Corticosteroids may be self-administered if in oral formulation and require minimal resources for administration in intravenous form. A short course of corticosteroids is likely to be affordable for the majority of patients if paying out-of-pocket.	The panel judged minimal barriers with regards to administration of systemic corticosteroids in the hospital setting, even in resource-constrained jurisdictions. The panel considered patient-level, provider-level and system-level factors that may serve as barriers for implementation of the recommendation. At the patient level, financial burden, difficulty with self-administration and adherence for long courses of treatment were considered, none of which were felt to have bearing in this context. At the provider level, no significant resources are required for administration in a hospital setting, even in resource-constrained jurisdictions. At the system level, the panel judged that the treatment is readily accessible and available. The panel therefore agreed that the treatment was feasible.

SUMMARY OF JUDGEMENTS							
	JUDGEMENT						
PROBLEM	No	Probably no	Probably yes	Yes		Varies	Don't know
DESIRABLE EFFECTS	Trivial	Small	Moderate	Large		Varies	Don't know
UNDESIRABLE EFFECTS	Trivial	Small	Moderate	Large		Varies	Don't know
CERTAINTY OF EVIDENCE	Very low	Low	Moderate	High			No included studies
VALUES	Important uncertainty or variability	Possibly important uncertainty or variability	Probably no important uncertainty or variability	No important uncertainty or variability			
BALANCE OF EFFECTS	Favors the comparison	Probably favors the comparison	Does not favor either the intervention or the comparison	Probably favors the intervention	Favors the intervention	Varies	Don't know
RESOURCES REQUIRED	Large costs	Moderate costs	Negligible costs and savings	Moderate savings	Large savings	Varies	Don't know

CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	Very low	Low	Moderate	High			No included studies
COST EFFECTIVENESS	Favors the comparison	Probably favors the comparison	Does not favor either the intervention or the comparison	Probably favors the intervention	Favors the intervention	Varies	No included studies
EQUITY	Reduced	Probably reduced	Probably no impact	Probably increased	Increased	Varies	Don't know
ACCEPTABILITY	No	Probably no	Probably yes	Yes		Varies	Don't know
FEASIBILITY	No	Probably no	Probably yes	Yes		Varies	Don't know

TYPE OF RECOMMENDATION

Strong recommendation against the intervention ○	Conditional recommendation against the intervention ○	Conditional recommendation for either the intervention or the comparison ○		Strong recommendation for the intervention ○
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CONCLUSIONS

Recommendation

In immunocompetent adults hospitalized with less severe community-acquired pneumonia, we suggest administering a seven-day course of systemic corticosteroids over no corticosteroids. (conditional or weak recommendation, low certainty evidence)

Justification

When moving from evidence to recommendations, the panel emphasized a large benefit observed in a reduced risk of needing invasive mechanical ventilation, and a moderately large benefit in increased ventilator-free days. The panel balanced these benefits against trivial risks of secondary infections and gastrointestinal bleeding, and a trivial effect on all-cause mortality. Overall certainty of evidence was low.

Selection of a seven-day regimen was guided by a non-linear dose-response relationship observed between corticosteroids and mortality (7). No mortality benefit was seen in patients with less severe CAP; nonetheless, the panel anticipated similar benefits (and dose-response relationship) applicable to other outcomes and agreed a seven-day course remained reasonable.

The panel anticipated that while the majority of patients will be inclined to receive corticosteroids given the balance of benefits and harms, a reasonable proportion may be disinclined from doing so in the absence of a clear survival benefit, residual possibility of adverse events and a 0.7% increased risk in all-cause mortality in absolute terms. The weak recommendation warrants engaging in shared decision-making with patients. Variability in values and preferences across patients is anticipated.

The panel anticipated no major issues with regards to resources required, feasibility and acceptability with regards to the recommendation to use systemic corticosteroids. Corticosteroids are easy to administer as an oral medication, widely available across most jurisdictions, inexpensive both in terms of out-of-pocket expenses incurred by patients and from a health systems perspective, and feasible to administer for a 7-day course in the in-patient setting. The recommendation to use systemic corticosteroids was anticipated to reduce, rather than increase, health equity issues, given widespread availability, and no plausible difference in treatment effect or disproportionate burden placed on disadvantaged groups relative to others.

Subgroup considerations

No important subgroup effects were identified based on need for intensive care admission and vasopressor requirements at baseline (7). The panel developed a recommendation applicable to all patients with less severe CAP, and did not identify any relevant subgroup considerations within this population based on available evidence.

Implementation considerations

No patient-, provider- or system-level barriers to implementation were anticipated.

Monitoring and evaluation

The effect of corticosteroids on secondary infections and gastrointestinal bleeding is trivial based on moderate to high certainty evidence. No specific monitoring or evaluation for adverse events is warranted.

Research priorities

Areas of uncertainty and warranting future research include:

1. Evidence regarding values and preferences of hospitalized patients with CAP, specifically regarding how patients weigh different outcomes, and how the balance of benefits and harms are perceived;
2. Core outcome data set for patients hospitalized with CAP;
3. Cost-effectiveness analysis incorporating all patient-important outcomes including ventilator support for patients hospitalized with less severe CAP;
4. Evaluation of the impact of corticosteroids in patients with CAP and concomitant chronic obstructive pulmonary disease; and
5. Evaluation of the impact of corticosteroids in patients with CAP and immunosuppressive conditions.

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The panel additionally reviewed a more detailed evidence profile, providing an overview of treatment effects across all critical and important benefit and harm outcomes:

Question: Short-term systemic corticosteroids (7 days) compared to no systemic corticosteroids for immunocompetent adults hospitalized with less severe community-acquired pneumonia

Setting: In-patient hospital setting (emergency department, medical ward, and critical care units)

Bibliography:

1. Pitre T, Abdali D, Chaudhuri D, Pastores SM, Nei AM, Annane D, et al. Corticosteroids in community-acquired bacterial pneumonia: a systematic review, pairwise and dose-response meta-analysis. J Gen Intern Med. 2023;1–14.
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Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies (n)	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	short-term systemic corticosteroids	no systemic corticosteroids	Relative (95% CI)	Absolute (95% CI)		
All-cause mortality (follow-up: 90 days)												
7	randomised trials	not serious	not serious	not serious ^{a,b}	very serious ^c	none	94/1092 (8.6%)	94/1154 (8.1%)	RR 1.08 (0.83 to 1.42)	7 more per 1,000 (from 14 fewer to 34 more)	⊕⊕○○ Low	CRITICAL
Need for invasive mechanical ventilation (follow-up: 90 days)												
9	randomised trials	not serious	not serious	not serious ^{a,d}	not serious ^e	none	70/1371 (5.1%)	116/1298 (8.9%)	RR 0.56 (0.42 to 0.74)	39 fewer per 1,000 (from 52 fewer to 23 fewer)	⊕⊕⊕⊕ High	CRITICAL
Ventilator-free days (follow-up: 90 days)												
2 (630)	randomised trials	not serious	not serious	not serious ^{f,g}	serious ^h	none	22.0 days	19.1 days	The mean ventilator-free days was 22.0 days.	MD 2.9 days more (0.95 more to 4.85 more)	⊕⊕⊕○ Moderate	CRITICAL
Duration of hospitalization (follow-up: 90 days)												
8 (2498)	randomised trials	not serious ⁱ	not serious ^j	not serious ^d	serious ^k	none	7.4 days	8.5 days	The mean duration of hospitalization was 7.8 days.	MD 1.1 days fewer (1.44 fewer to 0.72 fewer)	⊕⊕⊕○ Moderate	IMPORTANT
Time to symptom resolution (follow-up: 90 days)												
9 (1322)	randomised trials	not serious	not serious ^l	not serious ^m	serious ⁿ	none	4.2 days	6.0 days	The mean time to symptom resolution was 5.4 days.	MD 1.83 days fewer (2.45 fewer to 1.21 fewer)	⊕⊕⊕○ Moderate	IMPORTANT
Secondary infections (follow-up: 90 days)												
10	randomised trials	not serious	not serious	not serious ^o	serious ^p	none	125/1354 (9.2%)	115/1376 (8.4%)	RR 1.09 (0.85 to 1.41)	8 more per 1,000 (from 13 fewer to 34 more)	⊕⊕⊕○ Moderate	IMPORTANT
Gastrointestinal bleeding (follow-up: 90 days)												
11	randomised trials	not serious	not serious	not serious	not serious ^q	none	29/1667 (1.7%)	29/1643 (1.8%)	RR 0.95 (0.56 to 1.60)	1 fewer per 1,000 (from 8 fewer to 11 more)	⊕⊕⊕⊕ High	IMPORTANT

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

Explanations

- a. Systematic review authors acknowledged that included studies used heterogeneous definitions for disease severity. Nonetheless, the panel did not feel this justified rating down one level for indirectness in this context.
- b. Data presented is from subgroup analysis based on disease severity, focused on patients with less severe community-acquired pneumonia. Data specific to the subgroup is presented due to a credible subgroup of effect by disease severity for the outcome of mortality identified in a systematic review by Pitre and colleagues. Therefore, we did not rate down for indirectness here.
- c. Rated down twice for imprecision due to confidence intervals including the pre-defined minimally important difference threshold for both important benefit and important harm (certainty rated in a trivial effect).
- d. No subgroup effect was identified based on disease severity (more severe versus less severe disease). Pooled results combining both subgroups are therefore presented for this outcome, and the panel did not rate down for indirectness in the absence of a credible subgroup effect.
- e. Confidence intervals did not cross the pre-defined minimally important difference; certainty was rated in an important benefit. The panel therefore did not rate down for imprecision.
- f. Follow-up duration of interest among included studies in the systematic review by Pitre and colleagues was not clearly defined. We conservatively defined our follow-up duration of interest as being 90 days from time of diagnosis, consistent with the follow-up duration used for other outcomes.
- g. Outcome reported in the accompanying systematic review by Pitre and colleagues was ventilator-free days; duration of invasive mechanical ventilation was not reported. The panel considered both outcomes to be measuring the same construct, and therefore did not rate down for indirectness based on outcomes.
- h. Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference threshold for important versus trivial effects (i.e. mean difference of 1 day) when rating certainty in important benefit.
- i. Systematic review by Pitre and colleagues identified a statistically significant subgroup effect based on risk of bias, where trials with higher risk of bias showed substantially greater reductions in durations of hospitalization. The panel therefore only considered low risk of bias studies here, and did not rate down for risk of bias accordingly.
- j. Forest plots provided in the accompanying systematic review by Pitre and colleagues showed no significant heterogeneity in effect estimates between studies at low risk of bias and those at high risk of bias. The panel therefore did not rate down for inconsistency.
- k. Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference for important versus trivial effects (i.e. mean difference of 1 day) when rating certainty in important benefit.
- l. Systematic review authors noted significant statistical heterogeneity across pooled studies with no clear credible subgroup effects observed. However, on evaluation of the forest plot, significant overlap in central estimates and confidence intervals was observed. The panel therefore did not rate down for inconsistency.
- m. Time to clinical cure was informed by a Cochrane systematic review by Stern and colleagues conducted in 2017 (Stern A, Skalsky K, Avni T, Carrara E, Leibovici L, Paul M. Corticosteroids for pneumonia. Cochrane Database Syst Rev. 2017;(12):CD007720. doi: 10.1002/14651858.CD007720.pub3). Authors used the following definition: "at least by no fever, haemodynamic stability, and return to baseline respiratory condition; we accepted the study definitions for clinical cure, recorded them, and assessed their compatibility with the IDSA outcome definitions." The panel judged the reported outcome in the accompanying Cochrane review to be similar to the prioritized outcome of interest, and judged that both measured the same construct; the panel therefore did not rate down for indirectness in this context.
- n. Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference for important versus trivial effects (i.e. mean difference of 2 day) when rating certainty in a trivial benefit.
- o. Systematic review authors acknowledged that included studies used heterogeneous definitions for the outcome of interest. Nonetheless, the panel did not feel this justified rating down one level for indirectness, given evaluated outcomes measured the same construct.
- p. Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference for important versus trivial effects (i.e. 20 more events per 1000 patients) when rating certainty in a trivial harm.
- q. Confidence intervals did not cross the pre-defined minimally important difference; certainty was rated in a trivial harm. The panel therefore did not rate down for imprecision.

The panel completed a similar EtD framework and reviewed a similar Summary of Findings table in an alternate format produced using MAGICapp, an alternate platform for digitally-produced and disseminated, user-friendly practice guidance.

For immunocompetent adults hospitalized with less severe community-acquired pneumonia	
Weak recommendation	
We suggest administering a seven-day course of systemic corticosteroids over no corticosteroids (<i>conditional or weak recommendation, low certainty evidence</i>).	
Evidence to decision	
Benefits and harms	Substantial net benefits of the recommended alternative
A literature review was conducted using OVID Medline and Embase to identify recent systematic reviews published since January 2015 addressing the prioritized guideline question. Of the identified reviews, one by Pitre and colleagues, published in the Journal of General Internal Medicine in April 2023, was used to inform the majority of outcomes prioritized by the panel [7]. Systemic corticosteroids result in 39 fewer patients requiring invasive mechanical ventilation (relative risk reduction of 46%) (high certainty), and probably prolong ventilator-free days by 2.9 days over a 90-day follow-up (moderate certainty). Corticosteroids may have a trivial effect on mortality (low certainty). Treatment probably reduces duration of hospitalization and time to symptom resolution (both moderate certainty). Systemic corticosteroids probably have a trivial effect on secondary infection (moderate certainty) and have a trivial effect on gastrointestinal bleeding (high certainty). Corticosteroids may result in 7 more deaths per 1000 treated patients. Judgements regarding important versus trivial (unimportant effects) and related interpretations of effects: The panel pursued a stepwise process to establish minimal important difference (MID) thresholds for all critical and important outcomes of interest. After consulting clinical experts, they judged a systematic survey of studies evaluating patient values and preferences was likely to be low-yield, and conducting their own survey or focus groups with patients did not prove feasible. In the absence of such evidence, the panel relied on its own experiences engaging in shared decision-making with patients in judging the values and preferences of average patients, and accordingly their judgements regarding the threshold for important versus trivial effects. The panel adopted a consensus-based approach to establish MIDs, and established thresholds for small, moderate and large effects for each prioritized outcome (though rated certainty in presence of an important versus a trivial effect): • Mortality: 1% small but important effect, 1.5% moderate effect, 2% large effect. • Need for invasive mechanical ventilation: 1% small but important effect, 1.5% moderate effect, 2% large effect. • Ventilator-free days (mean difference): 1 day small but important effect, 2 days moderate effect, 3 days large effect. • Duration of hospitalization (mean difference): 1 day small but important effect, 2 days moderate effect, 3 days large effect. • Time to symptom resolution (mean difference): 2 days small but important effect, 3 days moderate effect, 4 days large effect. • Secondary infections: 2% small but important effect, 3% moderate effect, 4% large effect. • Gastrointestinal bleeding: 2% small but important effect, 3% moderate effect, 4% large effect. Accordingly, the panel viewed treatment effects as being small but important for duration of hospitalization and time to clinical cure. The panel agreed that the moderate to large effects evident for invasive ventilation-related outcomes were in keeping with an overall large benefit despite lack of a survival benefit. The panel viewed treatment effects on secondary infections and gastrointestinal bleeding as being trivial. The panel similarly interpreted effects on mortality as being trivial, given a MID of 10 deaths per 1000 patients; they did, however, recognize that a reasonable proportion of patients may view this absolute increase in risk as small but important. Harms were deemed trivial overall.	

Certainty of the evidence	Low
For critically important outcomes, certainty of evidence ranged from low (for all-cause mortality) to high (for need for invasive mechanical ventilation). Overall certainty was therefore deemed to be low, in keeping with the lowest certainty rating among critical outcomes. For five of seven outcomes, evidence was downgraded for imprecision due to wide confidence intervals encompassing possibility of both important benefit and trivial effects. Certainty in mortality was rated down twice for imprecision, given confidence intervals encompassed the possibility of both important benefit and important harm while rating certainty in a trivial effect.	
Values and preferences	Substantial variability is expected or uncertain
A review of the COMET database yielded no existing core outcome data sets for patients hospitalized with CAP. A literature review regarding how patients hospitalized with CAP value the prioritized outcomes was not identified; an additional search for indirect evidence expanding to other forms of pneumonia or respiratory illness was de-prioritized due to feasibility considerations. Possibly important uncertainty regarding patient values and preferences exists at two levels: one, the relative weight patients place on the prioritized outcomes; and two, the extent to which patients may value large benefits in avoiding invasive mechanical ventilation relative to trivial risk of harms including 90-day all-cause mortality, serious infections and gastrointestinal bleeding. The panel relied on its own judgments of what it perceived average well-informed patients would value when considering benefits, harms and burdens of treatment. The panel agreed that the majority of patients hospitalized with less severe CAP would be inclined to accept short-term therapy when faced with high certainty of moderately large or large benefit, and low certainty of trivial harm.	
Resources	Important issues, or potential issues not investigated
Resources required: Systemic corticosteroids are inexpensive, easy to administer, and are readily available globally. Dexamethasone and prednisone are among the most commonly listed drugs on the national essential medicines list, available for use in over 95% of countries internationally [8]. A seven-day course of corticosteroids has been shown to be an optimal regimen [7]. Dexamethasone 6mg tablets costs approximately \$1.78 - \$1.90 USD per tablet [9]; a 7-day course is estimated to cost a total of \$12.46-\$13.30 USD. While costs may vary somewhat across jurisdictions and based on available corticosteroid agents, dosages and formulations, the panel agreed they are likely to be negligible universally. The panel considered that a seven-day course of treatment is time-limited, and that the intervention is not associated with cumulative long-term costs. Panel members agreed that the cost of a one-week course of corticosteroids is likely to be affordable to most patients paying out-of-pocket. Moreover, the panel acknowledged that resources required, costs and barriers to access may be further mitigated in the in-patient setting, where corticosteroids are almost universally available, are often covered with governmental health insurance for in-patients, and require minimal resources to administer. Cost effectiveness: A literature review was conducted using PubMed with keywords "cost effectiveness", "corticosteroids" and "community acquired pneumonia" to identify studies evaluating cost effectiveness of systemic corticosteroids for community acquired pneumonia. One study by Pliakos and colleagues [10] evaluated the cost-effectiveness based on a decision analysis model which calculated deaths averted and incremental cost-effectiveness ratios with the addition of corticosteroids to antibiotics. Among hospitalized patients with Pneumonia Severity Index scores of I to III, corticosteroids resulted in additional costs of \$483,016 per death averted; only 29.9% of simulations demonstrated cost-effectiveness. Lack of cost-effectiveness for these patients was driven by very low mortality rates. No other recent cost-effectiveness analyses were identified. Given a lack of survival benefit (and instead, a signal towards possible increased mortality) with corticosteroids, the panel viewed the results of the stated cost-effectiveness analysis, which is based on deaths averted, as less applicable to the current context.	

In the absence of other cost-effectiveness analyses, the panel acknowledged an absence of reliable cost-effectiveness data to base its judgments on.

Equity

No important issues with the recommended alternative

No research evidence was identified pertaining to equity.

Systemic corticosteroids are anticipated to be accessible for hospitalized patients with less severe CAP across the vast majority of jurisdictions, including low and middle-income settings [8]. Where patients may pay out of pocket for treatment, systemic corticosteroids are anticipated to contribute minimally to the total cost of care and are anticipated to be accessible from a financial perspective for patients across the vast majority of financial contexts.

The panel judged that recommending the use of systemic corticosteroids was unlikely to reduce, and may instead possibly increase, health equity.

The panel did not identify plausible reasons for differences in the relative effectiveness of systemic corticosteroids for disadvantaged groups.

Acceptability

No important issues with the recommended alternative

No research evidence was identified pertaining to acceptability.

Systemic corticosteroids are easy to administer as an oral medication, and a short course of corticosteroids of 7 days is anticipated to be viewed favourably by the majority of patients.

The panel, including four patient partners, agreed that most patients hospitalized with less severe CAP would be inclined to accept short-term systemic corticosteroids when faced with moderate or high certainty of moderately large or large potential benefit; and low, moderate or high certainty of trivial risk of harms. The panel therefore judged that the intervention would be acceptable to all or almost all patients.

The panel also considered acceptability issues for health care providers involved in the administration of the medication. Systemic corticosteroids were not perceived to pose any major challenges in terms of drug administration.

Feasibility

No important issues with the recommended alternative

Systemic corticosteroids are widely available [8] and are easy to administer as oral medications. Treatment course is short and time-limited at 7 days. Corticosteroids may be self-administered if in oral formulation and require minimal resources for administration in intravenous form. A short course of corticosteroids is likely to be affordable for the majority of patients if paying out-of-pocket.

The panel judged minimal barriers with regards to administration of systemic corticosteroids in the hospital setting, even in resource-constrained jurisdictions.

The panel considered patient-level, provider-level and system-level factors that may serve as barriers for implementation of the recommendation. At the patient level, financial burden, difficulty with self-administration and adherence for long courses of treatment were considered, none of which were felt to have bearing in this context. At the provider level, no significant resources are required for administration in a hospital setting, even in resource-constrained jurisdictions. At the system level, the panel judged that the treatment is readily accessible and available.

The panel therefore agreed that the treatment was feasible.

Rationale

When moving from evidence to recommendations, the panel emphasized a large benefit observed in a reduced risk of needing invasive mechanical ventilation, and a moderately large benefit in increased ventilator-free days. The panel balanced these benefits against trivial risks of secondary infections and gastrointestinal bleeding, and a trivial effect on all-cause mortality. Overall certainty of evidence was low.

Selection of a seven-day regimen was guided by a non-linear dose-response relationship observed between corticosteroids and mortality [7]. No mortality benefit was seen in patients with less severe CAP; nonetheless, the panel anticipated similar benefits (and dose-response relationship) applicable to other outcomes and agreed a seven-day course remained reasonable.

The panel anticipated that while the majority of patients will be inclined to receive corticosteroids given the balance of benefits and harms, a reasonable proportion may be disinclined from doing so in the absence of a clear survival benefit, residual possibility of adverse events and a 0.7% increased risk in all-cause mortality in absolute terms. The weak recommendation warrants engaging in shared decision-making with patients. Variability in values and preferences across patients is anticipated.

The panel anticipated no major issues with regards to resources required, feasibility and acceptability with regards to the recommendation to use systemic corticosteroids. Corticosteroids are easy to administer as an oral medication, widely available across most jurisdictions, inexpensive both in terms of out-of-pocket expenses incurred by patients and from a health systems perspective, and feasible to administer for a 7-day course in the in-patient setting. The recommendation to use systemic corticosteroids was anticipated to reduce, rather than increase, health equity issues, given widespread availability, and no plausible difference in treatment effect or disproportionate burden placed on disadvantaged groups relative to others.

Practical info

The panel advised that systemic corticosteroids should be used for seven days.

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PICO (1.1) Population: Adults hospitalized with less severe community-acquired pneumonia Intervention: Systemic corticosteroids Comparator: No systemic corticosteroids					
Outcome Timeframe	Study results and measurements	Absolute effect estimates		Certainty of the evidence (Quality of evidence)	Summary
		No corticosteroids	Corticosteroids		
All-cause mortality 90 days	Relative risk: 1.08 (CI 95% 0.83 - 1.42) Based on data from 2246 participants in 7 studies	81 per 1000 Difference: 6 more per 1000 (CI 95% 14 fewer - 34 more)	87 per 1000	Low Due to very serious imprecision ¹	Systemic corticosteroids may have little or no effect on all-cause mortality.
Need for invasive mechanical ventilation 90 days	Relative risk: 0.56 (CI 95% 0.42 - 0.74) Based on data from 2669 participants in 9 studies	89 per 1000 Difference: 39 fewer per 1000 (CI 95% 52 fewer - 23 fewer)	50 per 1000	High ₂	Systemic corticosteroids reduce the risk of needing invasive mechanical ventilation.
Secondary infections 90 days	Relative risk: 1.09 (CI 95% 0.85 - 1.41) Based on data from 2730 participants in 10 studies	84 per 1000 Difference: 8 more per 1000 (CI 95% 13 fewer - 34 more)	92 per 1000	Moderate Due to serious imprecision ³	Systemic corticosteroids probably have little or no effect on secondary infections.
Gastrointestinal bleeding 90 days	Relative risk: 0.95 (CI 95% 0.56 - 1.6) Based on data from 3310 participants in 11 studies	18 per 1000 Difference: 1 fewer per 1000 (CI 95% 8 fewer - 11 more)	17 per 1000	High ₄	Systemic corticosteroids have little or no effect on gastrointestinal bleeding.
Ventilator-free days 90 days	Measured by: Scale: - High better Based on data from 630 participants in 2 studies	19.1 Mean Difference: MD 2.9 more (CI 95% 0.9 more - 4.9 more)	22.0 Mean	Moderate Due to serious imprecision ⁵	Systemic corticosteroids probably increase ventilator-free days.
Duration of hospitalization 90 days	Measured by: Scale: - Lower better Based on data from 2498 participants in 8 studies	8.5 Mean Difference: MD 1.1 fewer (CI 95% 1.4 more - 0.7 fewer)	7.4 Mean	Moderate Due to serious imprecision ⁶	Systemic corticosteroids probably reduce duration of hospitalization.
Time to symptom resolution 90 days	Measured by: Scale: - Lower better Based on data from 1322 participants in 9 studies	6.0 Mean Difference: MD 1.8 fewer (CI 95% 2.4 fewer - 1.2 fewer)	4.2 Mean	Moderate Due to serious imprecision ⁷	Systemic corticosteroids probably have little or no effect on time to symptom resolution.
1. Indirectness: no serious. 1) Systematic review authors acknowledged that included studies used heterogeneous definitions for disease severity. Nonetheless, the panel did not feel this justified rating down one level for indirectness in this context. 2) Data presented is from subgroup analysis based on disease severity, focused on patients with less severe community-acquired pneumonia. Data specific to the subgroup is presented due to a credible subgroup of effect by disease severity for the outcome of mortality identified in a systematic review by Pitre and colleagues. Therefore, we did not rate down for indirectness here.; Imprecision: very serious. Rated down twice for imprecision due to confidence intervals including the pre-defined minimally important difference threshold for both important benefit and important harm (certainty rated in a trivial effect).					

2. **Indirectness: no serious.** 1) Systematic review authors acknowledged that included studies used heterogeneous definitions for disease severity. Nonetheless, the panel did not feel this justified rating down one level for indirectness in this context. 2) No subgroup effect was identified based on disease severity (more severe versus less severe disease). Pooled results combining both subgroups are therefore presented for this outcome, and the panel did not rate down for indirectness in the absence of a credible subgroup effect.; **Imprecision: no serious.** Confidence intervals did not cross the pre-defined minimally important difference; certainty was rated in an important benefit. The panel therefore did not rate down for imprecision.
3. **Indirectness: no serious.** Systematic review authors acknowledged that included studies used heterogeneous definitions for the outcome of interest. Nonetheless, the panel did not feel this justified rating down one level for indirectness, given evaluated outcomes measured the same construct.; **Imprecision: serious.** Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference for important versus trivial effects (i.e. 20 more events per 1000 patients) when rating certainty in a trivial harm
4. **Imprecision: no serious.** Confidence intervals did not cross the pre-defined minimally important difference; certainty was rated in a trivial harm. The panel therefore did not rate down for imprecision.
5. **Indirectness: no serious.** 1) Follow-up duration of interest among included studies in the systematic review by Pitre and colleagues was not clearly defined. We conservatively defined our follow-up duration of interest as being 90 days from time of diagnosis, consistent with the follow-up duration used for other outcomes. 2) Outcome reported in the accompanying systematic review by Pitre and colleagues was ventilator-free days; duration of invasive mechanical ventilation was not reported. The panel considered both outcomes to be measuring the same construct, and therefore did not rate down for indirectness based on outcomes.; **Imprecision: serious.** Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference threshold for important versus trivial effects (i.e. mean difference of 1 day) when rating certainty in important benefit.
6. **Risk of Bias: no serious.** Systematic review by Pitre and colleagues identified a statistically significant subgroup effect based on risk of bias, where trials with higher risk of bias showed substantially greater reductions in durations of hospitalization. The panel therefore only considered low risk of bias studies here, and did not rate down for risk of bias accordingly.; **Inconsistency: no serious.** Forest plots provided in the accompanying systematic review by Pitre and colleagues showed no significant heterogeneity in effect estimates between studies at low risk of bias and those at high risk of bias. The panel therefore did not rate down for inconsistency.; **Indirectness: no serious.** No subgroup effect was identified based on disease severity (more severe versus less severe disease). Pooled results combining both subgroups are therefore presented for this outcome, and the panel did not rate down for indirectness in the absence of a credible subgroup effect.; **Imprecision: serious.** Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference for important versus trivial effects (i.e. mean difference of 1 day) when rating certainty in important benefit.
7. **Inconsistency: no serious.** Systematic review authors noted significant statistical heterogeneity across pooled studies with no clear credible subgroup effects observed. However, on evaluation of the forest plot, significant overlap in central estimates and confidence intervals was observed. The panel therefore did not rate down for inconsistency.; **Indirectness: no serious.** Time to clinical cure was informed by a Cochrane systematic review by Stern and colleagues conducted in 2017 (Stern A, Skalsky K, Avni T, Carrara E, Leibovici L, Paul M. Corticosteroids for pneumonia. Cochrane Database Syst Rev. 2017;(12):CD007720. doi: 10.1002/14651858.CD007720.pub3). Authors used the following definition: "at least by no fever, haemodynamic stability, and return to baseline respiratory condition; we accepted the study definitions for clinical cure, recorded them, and assessed their compatibility with the IDSA outcome definitions." The panel judged the reported outcome in the accompanying Cochrane review to be similar to the prioritized outcome of interest, and judged that both measured the same construct; the panel therefore did not rate down for indirectness in this context.; **Imprecision: serious.** Rated down once for imprecision due to confidence intervals including the pre-defined minimally important difference for important versus trivial effects (i.e. mean difference of 2 day) when rating certainty in a trivial benefit.

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1. Pitre T, Abdali D, Chaudhuri D, et al. Corticosteroids in Community-Acquired Bacterial Pneumonia: a Systematic Review, Pairwise and Dose-Response Meta-Analysis. *J Gen Intern Med* 2023;38(11):2593-606. doi: 10.1007/s11606-023-08203-6 [published Online First: 20230419]